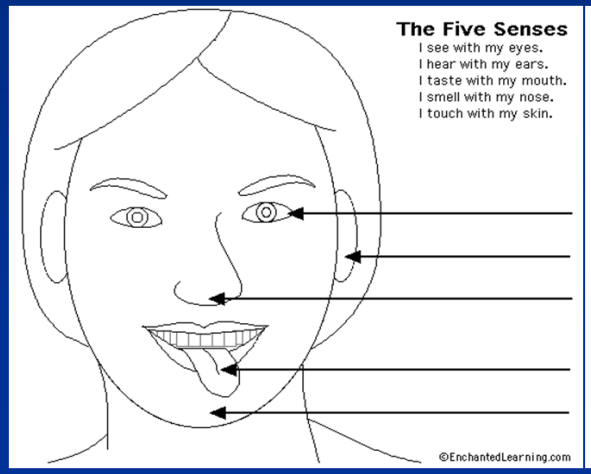
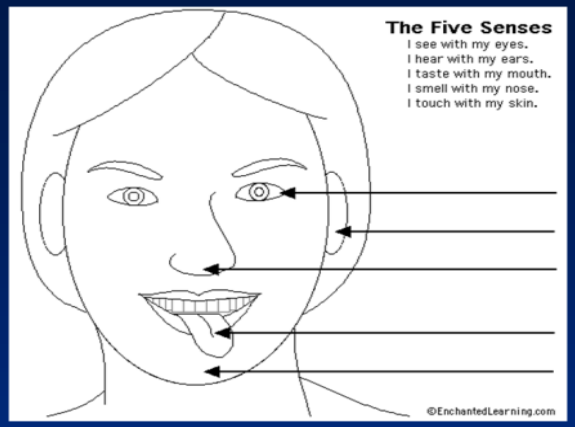
Chapter Three

Sensation and Perception





Sensation



Sensation

- is the simple experience that

  arises from:

  - The ***stimulation*** of the sense organs.

- It is the detection of ***physical energy***:

  - Emitted or
  - Reflected

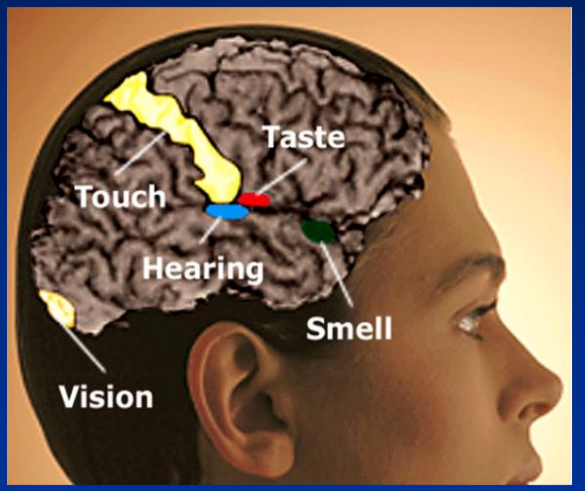
  } by ***physical objects***.



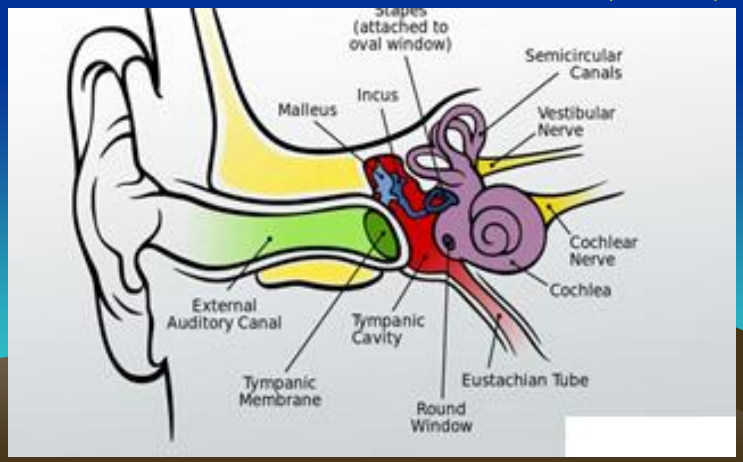
- Sensation is our first awareness of some outside stimulus.

Cont …

- It is the conversion of energy from:
  - The environment into a pattern of response by the ***nervous system***



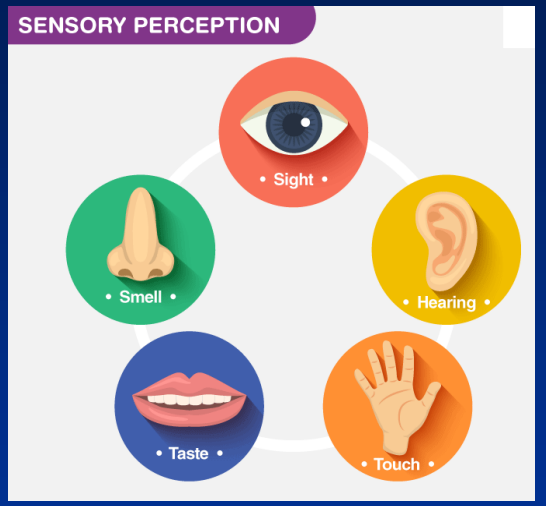
- The vestibular (ear) sense informs us about:
  - The movement and/or
  - Stationery position of the head.



Cont…

- The ***skeletal muscles*** contain receptors (called kinesthesis)
  - Responsible for a sense of:

- Bodily movement





- In general, sensory experiences contribute immeasurably to the ***QUALITY OF LIFE***

Cont…

Measuring Senses



- ***Absolute threshold***: is the smallest amount of energy that a person can detect reliably
- Reliable detection is said to occur when a person can detect a signal ***50 percent of the time***

➡ If you have **normal sensory abilities,** you can:



Absolute Threshold

The weakest amount of a stimulus that a person can detect 50% of the time.

Sight		Seeing a candle flame 30 miles away on a clear night
Hearing		Hearing a watch ticking 20 feet away
Touch		Feeling a bee's wing falling a distance of 1cm onto your cheek
Smell		Smelling one drop of perfume in a three room house
Taste		Tasting one teaspoon of sugar dissolved in two gallons of water

Cont…

- **Difference thresholds:**
  - is the ***smallest*** difference in stimulation that a person can ***detect reliably*** (again, half of the time)

For example, a person might be asked to compare:

- The ***weight*** of two blocks,





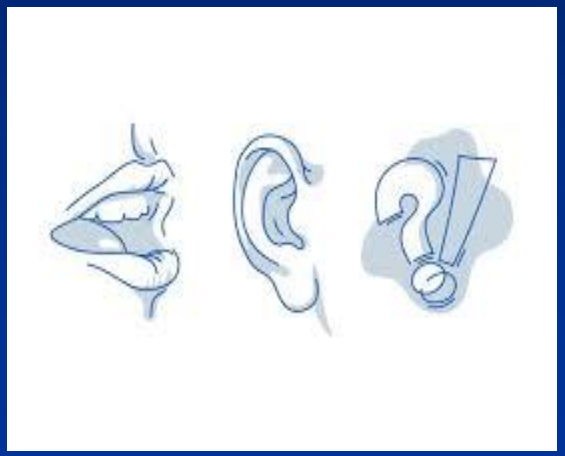
- The ***brightness*** of two lights or

- The ***saltiness*** of two fluids



Cont…

- **Sensory adaptation:** When certain stimulus is ***unchanged*** or ***repeated*** for a long period of time,



sensation often becomes ***fade*** or ***disappear***



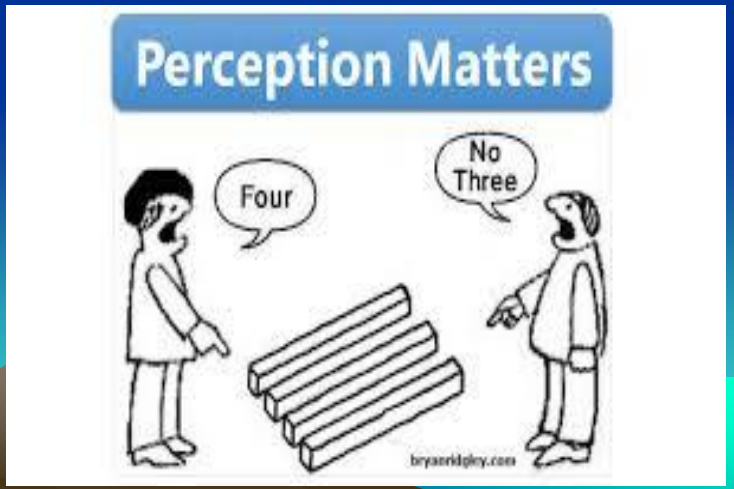


The Meaning of Perception



- Perception is the cognitive process of:
  - ***Selecting***, ***organizing*** and ***interpreting*** stimuli/sensory input to understand the world.
  - It is the ***interpretation of that information***.

Example: Detecting whether there are black lines on a white sheet of paper is **sensation** and **Identifying** them as l***etter, images*** and ***further*** more understanding is called ***perception***



Cont …

Visual Perception:



- In the brain, sensory signals that give rise to:
  - ***Vision, hearing, taste,*** } are ***combined*** produce
  - ***Smell, and touch*** } a ***unified model of world***
- Perception consists of ***three basic processes***:
  - ***Selection*****:** the process ***selecting*** a stimuli to attend while ***disregarding the rest*** – ***Selective attention***
  - ***Organization*****:** ***assembling*** the selected sensations into usual patterns
  - ***Interpretation***: explaining the ***selected*** and ***organized*** sensations to make ***reasonable judgments*** from them.

Cont…

- **Selective attention is affected by: Env'tal and personal factors**
  - **Environmental Factors:**
    - **Intensity-** The **louder the sound, the brighter the light, the more it captures our attention**
    - **Size - We tend to notice usually large or small things**
    - **Contrast - What contrasts with the surrounding env't attracts attention easily.**
    - **Repetition - repetition of a stimulus is important to gain attention**

Cont…

- **Movement – Our eyes are involuntarily attracted to movement**
- **Novelty- we are attracted to sth. New in the env't**

- **Personal Factors:**
  - **Motivation** – our needs & desires influence our perception
  - **Belief -** what we believe to be true about the world affects our attention.
  - **Emotion -** A small child afraid of the dark

Example: He may see a ghost instead of a rope hanging on the door

Cont…

- **Experience -** Previous experiences often affect how we perceive the world

  - The tendency to perceive what to expect is called perceptual set.

Example: hearing a certain lines in a song may remind you…. While others are continuing their discussion

- **A mental predisposition to Perceive one thing & not another – perceptual set**

- What you see in the center is influenced by perceptual set

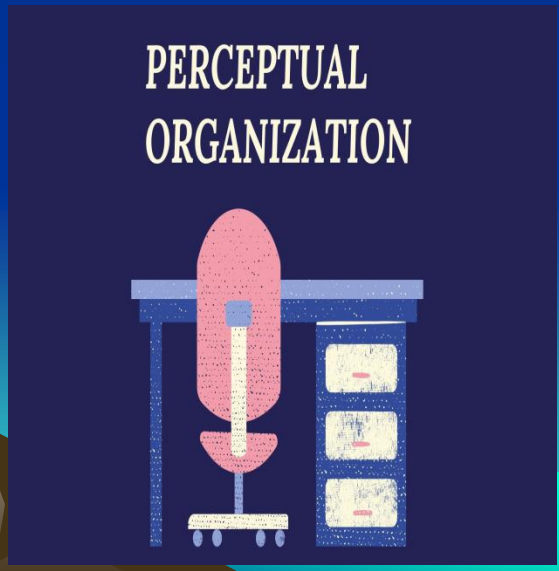


Cont…

- **Organization is** the process of:
  - ***Assembling*** the selected sensations into usual patterns to understand the world.
  - Organization of data can be divided in to:

A. **Perceptual Organization**

✓ **Gestalt psychologists are the first to study how people organize the world in to meaningful units or patterns**

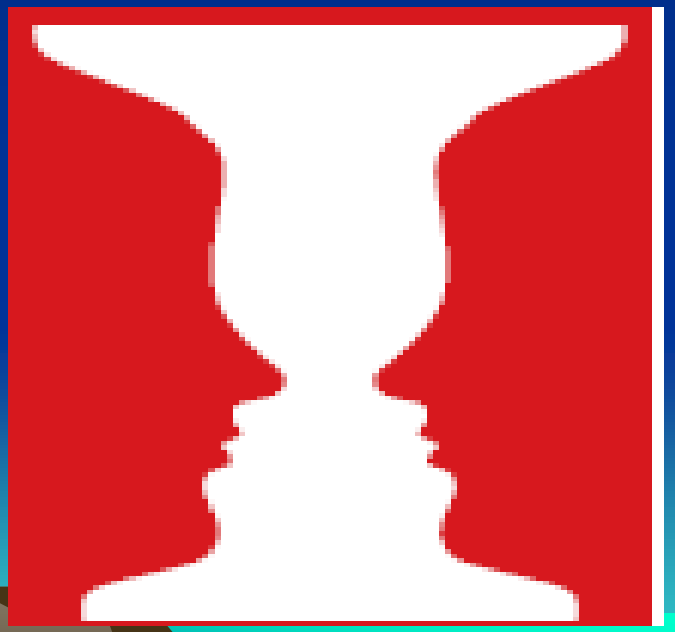


Cont…

- **They formulated a series of principles:**
  - **The figure ground principle:**

- In organizing stimuli, we tend to automatically distinguish between a figure and a ground

- The figure, with more detail, stands out against the background, which has less detail.



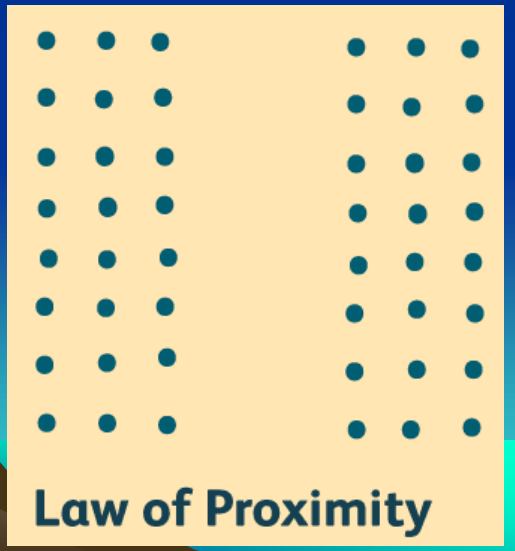
Cont…

- **The principle of perceptual grouping is:**

  - The tendency to perceive stimuli as meaningful wholes or patterns

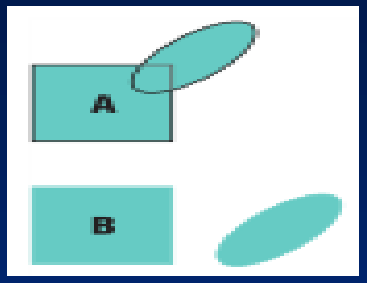
  - We see complex patterns of stimuli as unitary forms or objects by grouping on the basis of:

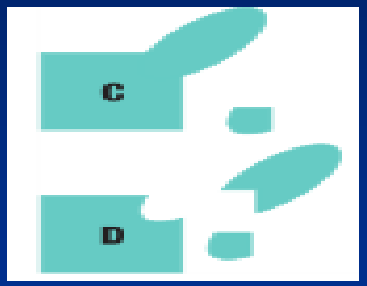
- **Proximity:** we group together ***objects that are physically close to one another.***



Cont…

- **Simplicity:** states that stimuli are **organized in the simplest way possible.**

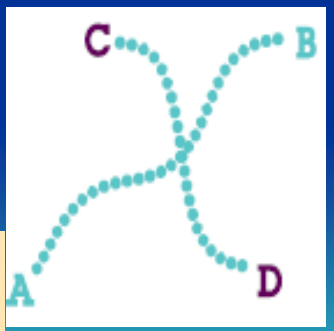




- This rule says that we tend to ***perceive complex figures*** as divided into ***several simpler figures***

- Simplest organization requires least cognitive effort

- **Continuity:** we tend to ***favour smooth or continuous paths when interpreting a series of points or lines.***

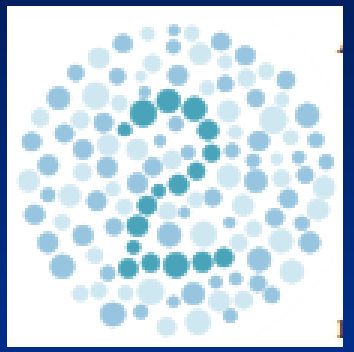




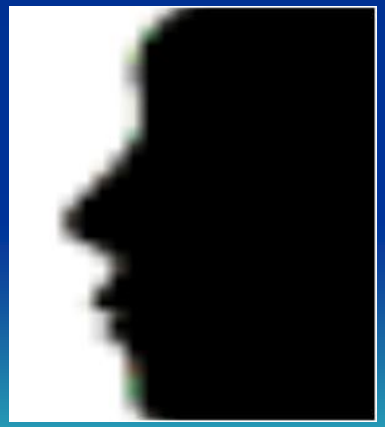
Cont…

- **Similarity:** Objects that are alike in some way (color, size or shape) tend to be perceived as belonging together.





- **Contour principle:** The boundary of a perceptual figure which gives shape to the objects in our visual field.

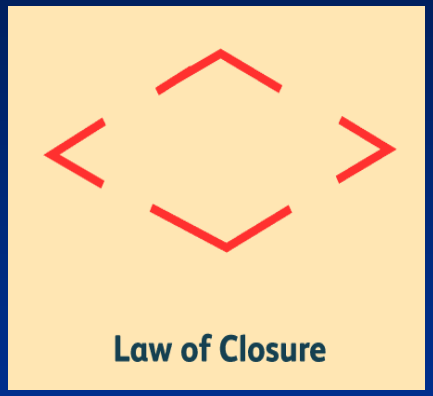




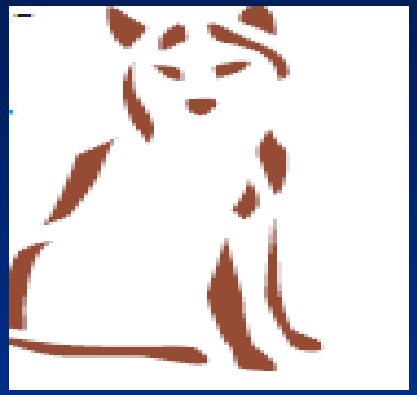
Example, provide ***shape*** for objects which have no shape

Cont…

- **Closure principle:** The brain tends to fill in

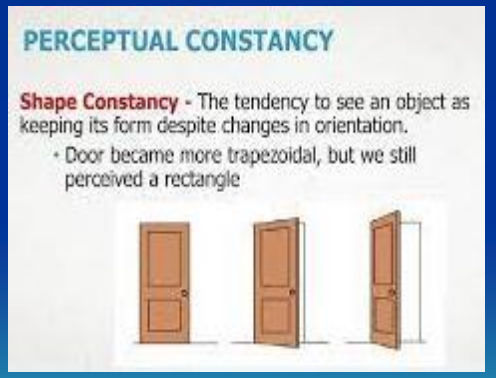


gaps in order to



perceive complete forms

B. **Perceptual constancy**



- Our tendency to perceive sizes, shapes, brightness, and colours as remaining the same even though their physical characteristics are constantly changing

Types of perceptual constancies

- **Shape constancy:** tendency to perceive an object as



retaining its shape even though its shape is continually changing its image on the retina.



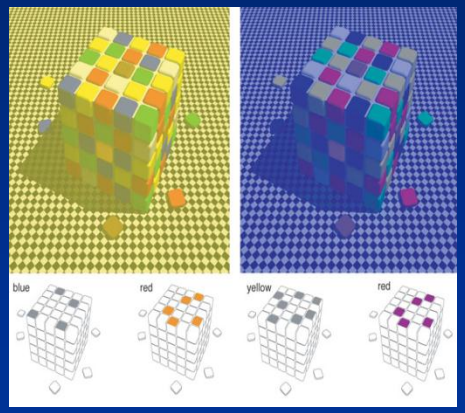
- **Size constancy**: tendency to perceive objects as remaining the ***same size*** as we get closer to or far from it..



Cont …



- **Brightness** constancy: we continue to see objects as having constant brightness though the overall illumination changes.



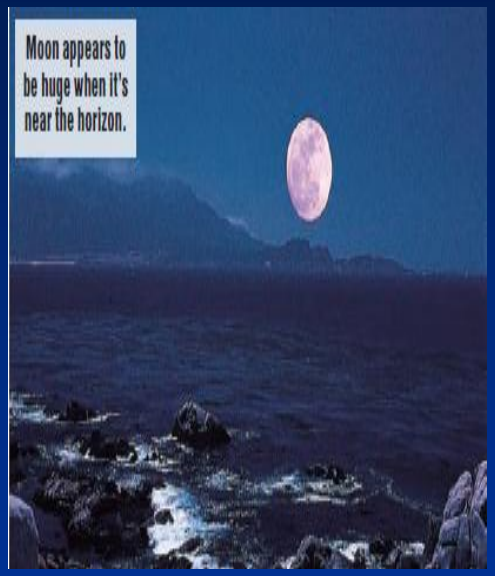
- **Colour** constancy: the tendency to perceive colours as having stability despite the d/ces in lighting.

- **Location** constancy: We perceive stationery objects as ***remaining in the same place***, even though the ***retinal image moves about as we move our eyes***

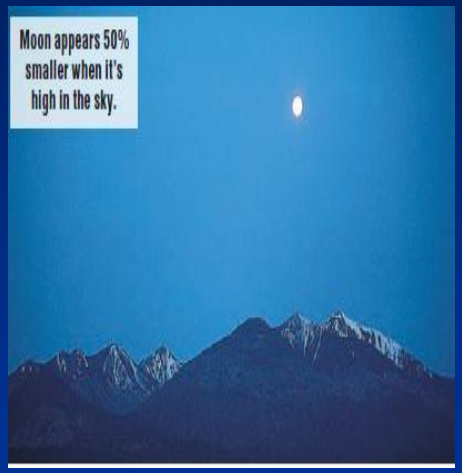
Example: As we drive…telephone poles and trees fly by- our retina…

Cont…

Visual Illusions:



- Illusion refers to instances in which perception yields false interpretation about the physical reality.



- Perceiving an image as being so ***strangely distorted*** that, in reality, it cannot and does not exist.
- Occurs when the brain can ***no longer correctly interpret space, size***, and ***depth*** cues

Depth and Distance Perception

- Refers to our remarkable ability to judge accurately how far objects are from us or from each other
- To perform this, we rely on:

  - **Binocular** cues: that require the use of the ***two eyes***

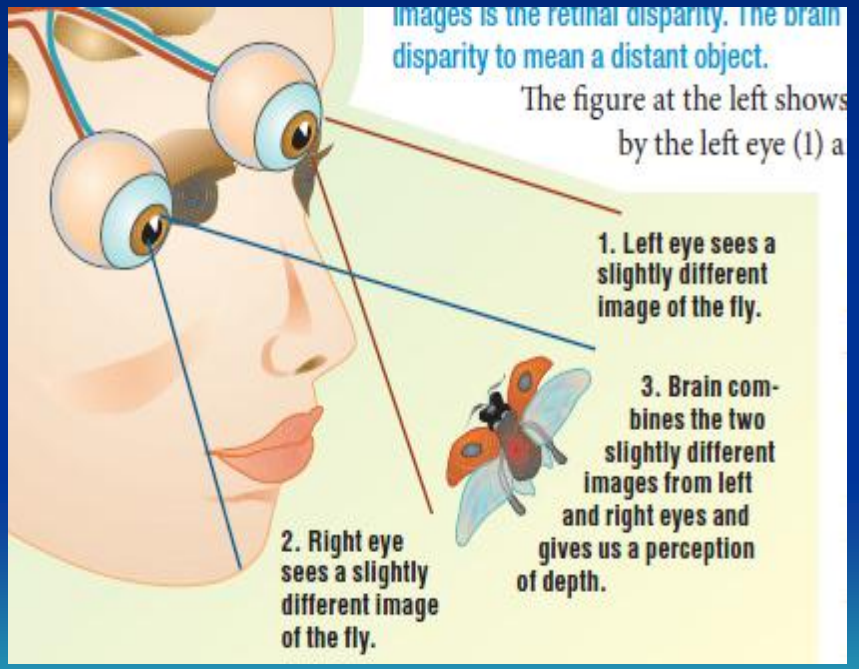
    - Convergence: the turning of the eyes inward, which occurs when they ***focus on nearby object***.



Cont …

- **Retinal disparity** refers to a binocular depth cue that depends on the distance b/n the eyes.

  - The difference between the right and left eyes' images is the retinal disparity.



Cont…

- **Monocular cues**: cues that **do not** depend on using both eyes.

  i.e signals produced by a single eye.

- They commonly arise from the way objects are arranged in the environment

➢ Some of the clues include:

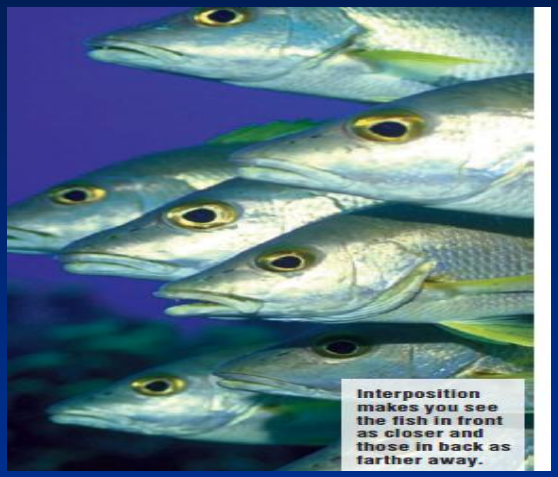
✓ **Linear perspective**: When two lines seem to be parallel appearing to come together or converge, they imply the existence of depth.



Cont…

- ✓ **Interposition**: When one object obstructs the view of another object /overlaps/,

  - The object that is in view appears closer than the other



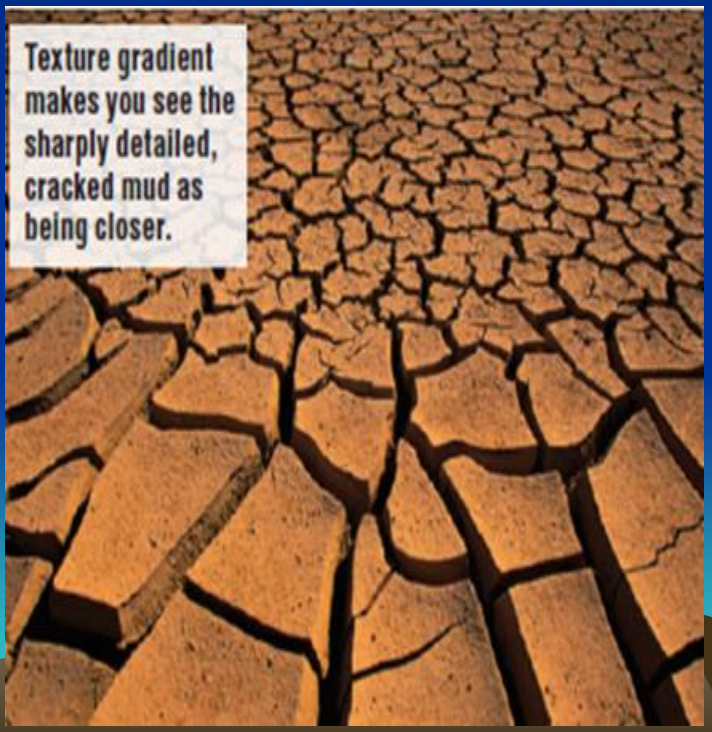
- ✓ **Aerial perspectives**: Objects that are far away look ***fuzzy and blurred*** ***in*** comparison to ***near objects b/c*** of intervening particles of dust, haze, or smoke in the atmosphere.



Cont…



- ✓ **Light and shadow**: ***Brighter objects appear closer***, while darker or dimmer objects are perceived as farther away..



- ✓ **Texture gradient**: Closer objects appear to have a ***rough or detailed*** texture. as distance increases, the texture appears become finer and finer

Cont…

- ✓ **Motion/Movement perception**

  - Real Motion: is the result of actual change of objects in space. There are **two** ways

    1) an *image moves across the retina*, and

    2) the *eye moves in the head* to follow the path of the moving object.

  - Apparent Motion: perceiving movement in the absence of any real motion.

    - ❑ Some of them are:-

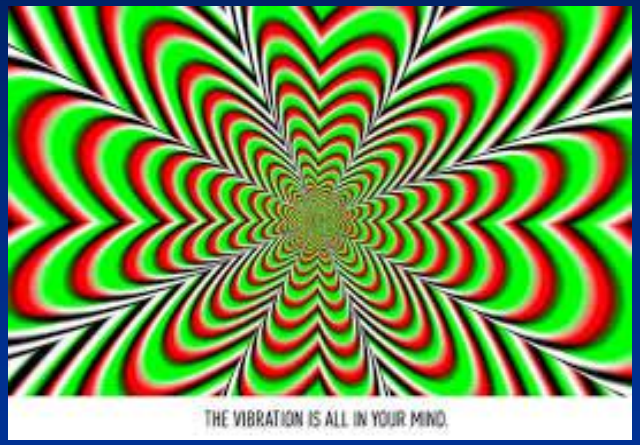
Cont…

- ✓ Stroboscopic motion: If you are seated in a darkened room and look at two adjacent lights being turned on, one after the other, it will seem that ***a single light is jumping back and forth***



Cont…

- ✓ Auto kinetic motion: perceived ***motion of a single stationer light or object.*** (A glowing end of cigarette)





- ✓ Induced movement: occurs if a stationery spot or object is perceived as moving when its frame or background moves.

E.g The movement of the moon



Extra sensory perception

- ESP is a group of psychic experiences that involve perceiving or sending information out side the normal sensory process. Types of ESP:-

- Telepathy: is a direct communication from ***one mind to another without the usual visual, auditory*** and other sensory signals.



Cont …

- Clairvoyance: is the ability to perceive events or objects that are ***out of sight***.



E.g. if sbd. is dead



- Psycho kinesis: is the ability to exert mind over matter –

E.g., by moving objects without touching them.

- Precognition: is the perception of an event that has not yet happened.

